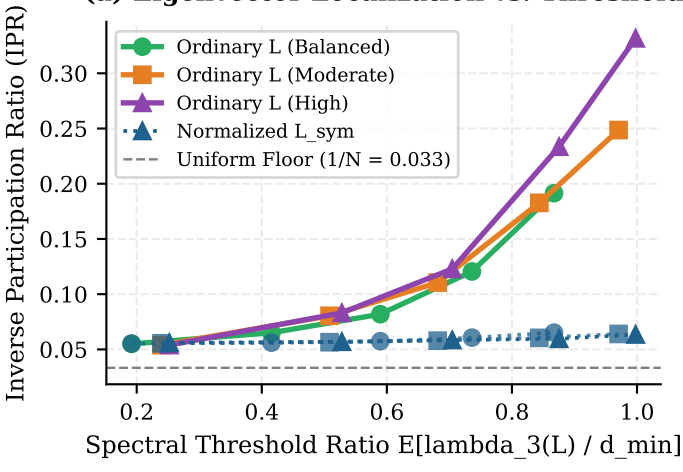
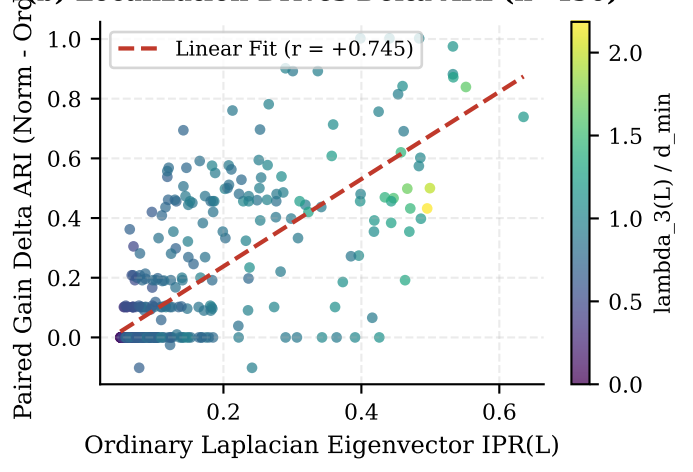

(a) Eigenvector Localization vs. Threshold
Ordinary L (Balanced)
Ordinary L (Moderate)
Ordinary L (High)
Normalized L_sym
Uniform Floor (1/N = 0.033)
Inverse Participation Ratio (IPR)
0.30
0.25
0.20
0.15
0.10
0.05
0.2
0.4
0.6
0.8
1.0
Spectral Threshold Ratio E[lambda_3(L) / d_min]


(b) Localization Drives Delta ARI (n=450)
Linear Fit (r = +0.745)
Paired Gain Delta ARI (Norm - Ord)
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
Ordinary Laplacian Eigenvector IPR(L)
lambda_3(L) / d_min
2.0
1.5
1.0
0.5
0.0